

PEDAGOGY OF

MATHEMATICS - I

B.Ed. - First Semester

Prepared under the New Syllabus of
TNTE University for B.Ed. Course



Prof. K. Aslam, M.Sc., M.Ed., M.Phil.,(Edu).,

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*Prepared under the new syllabus of
2021-2022, Tamilnadu Teacher Education
University*

Prof. K. ASLAM, M.Sc., M.Ed., M.Phil.,

EVEREST PUBLISHERS

Pearl Nest Apartment,

5/A, Thirunagar 1st Street,

Vadapalani,

Chennai – 600 026.

Price - ₹ 220/-

Topic of the Book	:	Pedagogy of Mathematics - I
Author	:	Prof. K. Aslam, M.Sc., M.Ed., M.Phil.
Edition	:	First Edition - January 2022
Copy right to	:	Everest Publishers
Publisher	:	Everest Publishers, Chennai - 26.
Printed at	:	Globe Printers, Chennai - 94.
Price	:	₹ 220/-

Copies can be available at the following address.

EVEREST PUBLISHERS

Pearl Nest Apartment,
5/A, Thirunagar 1st Street,
Vadapalani,
Chennai - 600 026.
Cell No : 9894472790 / 044-42070507

BRANCH AT COIMBATORE

EVEREST PUBLISHERS

J.K. Garden First Street,
Ukkadam,
Coimbatore - 641 001.
Cell No : 9894472790 / 9488551302

PREFACE

Pedagogy of mathematics - I is a book for B.Ed., first semester course of Tamilnadu Teacher Education university, Chennai, as to the new syllabus of 2021 - 22. All the content of Five units of the syllabus is briefly analysed and written for the understanding of all type of students in an easy manner. Necessary examples are taken from the Tamilnadu Text Books and is explained thoroughly.

As a professor of College of Education and having teaching experience more than thirty years. I have prepared this book in a short duration. At the end of each unit review questions are given as to the model of the university question pattern. In this book I have taken few notes from eminent authors and few internet informations and added it in this book. For this I convey my sincere thanks to them.

After reading this book, please convey your views and suggestions for the positive improvement of this book. I wish your goodself to encourage and motivate to fulfil the aim of this book.

K. ASLAM

E-mail: akaslam56@gmail.com

Cell : 9894472790

Chennai,
02-01-2022

PEDAGOGY OF MATHEMATICS

University Syllabus – 2021 - 2022 onwards

Unit-I : Aims and objectives of Teaching Mathematics

Meaning, Nature, Scope, Need and Significance, Values, Aims and Objectives: Instructional objectives and Behavioral Objectives – Need and Importance of Instructional Objectives. Bloom's Taxonomy of Instructional Objectives: Cognitive, Affective and Psychomotor Domains, Revised Bloom's Taxonomy 2001 (Anderson & Krathwohl) Interrelation among the domains – Correlation between subjects.

Unit-II : Teaching Skills

Micro-Teaching: Concept, Definition, Steps, Cycle, Micro-teaching Vs Macro-Teaching - Micro Teaching Skills: Skill of Set Induction, Skill of Explaining, Skill of Blackboard Usage, Skill of Questioning, Probing skills, Skill of Stimulus Variation, Skill of Reinforcement, Skill of non-verbal clues, Skill of Closure - Link lesson – Model episode.

Unit – III : Approaches of Teaching

Approaches of Lesson Planning - Steps - Organizing Teaching: Memory Level (Herbartian Model), Understanding Level (Morrison teaching Model), Reflective Level (Bigge and Hunt Teaching Model)– Unit Plan – Lesson Plan Writing.

Unit - IV : Methods of Teaching

Teacher Centered Instruction: Lecture method, Demonstration and Team Teaching – Learner Centered Instruction: Self-Learning – Forms of Self-Learning: Programmed Instruction, Computer Assisted Instruction, Keller Plan, Project Method, Activity Based Learning (ABL), Active Learning Method (ALM)-Mind Map, Advanced Active Learning Method (AALM).

Unit-V: Instructional Media

Classification of Instructional Media – Use of Mass media in classroom Instruction. New Emerging Media: Tele-Conferencing, Communication Satellites, Computer Networking, Word Processors, Blended Learning, Flipped Classroom, Artificial Intelligence, Augmented Reality

PEDAGOGY OF MATHEMATICS - I



CONTENTS

Units	Topic	Page No.
I	Aims and Objectives of Teaching Mathematics	
	1.1 Mathematics - Meaning	11
	1.2 National of Mathematics	13
	1.3 Scope of Mathematics	16
	1.4 Need and Significance Teaching Mathematics	17
	1.5 Values of Teaching Mathematics	18
	1.6 Aims and Objectives of Teaching Mathematics	21
	1.6.1. Aims and Objectives of Teaching Mathematics in Schools	26
	1.6.2. Difference between Aims and Objectives	30
	1.7 Instructional objectives and Behavioural objectives	30
	1.7.1. Writing Instructional objectives in terms of Behavioural objectives	35
	1.8 Blooms Taxonomy of Instructional Objectives	37
	1.8.1. Cognitive Domain	37
	1.8.2. Affective Domain	41
	1.8.3. Psychomotor Domain	44
	1.9 Revised Bloom's Taxonomy	45
	1.10 Writing Instructional Objectives for different Mathematics Topics	51
	1.11 Inter-relation among the domains	55
	1.12 Correlation between Mathematics and other subjects	58

Units	Topic	Page No.
II	Teaching skills	
	2.1 Micro Teaching	
	2.1.1. Micro Teaching - Definition	63
	2.1.2. Need and Importance of Micro teaching	63
	2.1.3. Characteristics of Micro teaching	64
	2.1.4. Micro Teaching cycle	65
	2.1.5. Uses of Micro Teaching	66
	2.1.6. Micro Teaching Vs. Macro Teaching	67
	2.2 Identification of teaching skills	69
	3.2.1. Skill of Set induction	71
	3.2.2. Skill of Explaining	74
	3.2.3. Skill of Black Board usage	75
	3.2.4. Skill of questioning	77
	3.2.5. Skill of probing	80
	3.2.6. Skill of Stimulus variation	82
	3.2.7. Skill of Reinforcement	84
	3.2.8. Skill of Non-verbal cues	85
	3.2.9. Skill of closure	87
	3.2.10. Advantages of micro teaching	89
	2.3 Model Micro Lessons	89
	2.4 Link Lesson	98
III	Approaches of Teaching	113
	3.1 Steps in planning a lesson	114
	3.2 Designing a Lesson Plan	117
	3.2.1. Herbartian Steps	118
	3.2.2. Preparation of Lesson Plan	120
	3.3 Organising Teaching	120
	3.3.1. Memory Level of Teaching	125
	3.3.2. Understanding Level of Teaching	128
	3.3.3. Reflective Level of Teaching	132
	3.4 Designing a Unit plan	137
	3.4.1. Difference between Lesson plan and unit plan	138
	3.5 Structure of a four-fold lesson plan	142
	3.6 Preparation of a model lesson plans	

Units	Topic	Page No.
IV	Methods of Teaching English	
	4.1 Teacher Centred methods	163
	4.1.1 Lector Method	163
	4.1.2 Demonstration Method	166
	4.1.3 Team Teaching	169
	4.2 Learner-Centred methods	171
	4.2.1 Self-Learning	172
	4.2.2 Programmed Instruction	175
	4.2.3 Computer Assisted Instruction	178
	4.2.4 The Keller Plan	180
	4.2.5 Project Method	183
	4.2.6 Activity Based Learning	187
	4.2.7 Active Learning method	191
	4.2.8 Mind map	195
	4.2.9 Advanced Active Learning method	199
V	Instructional Media	
	5.1 Use of Mass media for classroom instruction	203
	5.2 New Emerging Media	212
	5.2.1. Tele-conferencing	213
	5.2.2. Communication Satellites	216
	5.2.3. Computer Networking	218
	5.2.4. Word Processors	222
	5.2.5. Blended Learning	225
	5.2.6. Flipped Classroom	227
	5.2.7. Artificial Intelligence	231
	5.2.8. Augmented Reality	235

UNIT - 1

AIMS AND OBJECTIVES OF TEACHING MATHEMATICS

1.1 Mathematics : Meaning

The word Mathematics comes from the Greak Language which means, "that which is learnt" or "what one gets to know".

Aristotle defined mathematics as "the science of quantity".

Becon said, "Mathematics is the gateway and key to all sciences".

Loake said, "Mathematics is a way to settle in the mind a habit of reasoning".

Gause referred to Mathematics as "the Queen of sciences". We are seem to know what is mathematics but when it comes to define it, we seem to fail completely. A dictionary meaning of mathematics could be a science dealing with study of quantities and their relationships expressed in numbers and other special symbols. Mathematics is the study of topics such as quantity, structure, space and change (i.e. arithmetic, algebra, geometry and analysis)

Mathematics is defines as a Science of Abstract things. With the help of trigonometry we can find the hidden measurements. For example the distance between earth and moon, the distance between earth and son, distance between satellites etc. are obtained precisely.

Few parts of Mathematics subjects

1. **Number system** : This part is related to numbers and four basic operations of numbers.
2. **Algebra** : This section deals with numbers and unknown values ($x, y, z, \dots$). With the help of Algebraic expressions, Algebraic equations, we can find the unknown values.
3. **Geometry** : This deals with different figures or objects and their properties. This is related with point line, angle, lateral surface and concrete objects.
4. **Trigonometry** : It explains the correlation between angled and their sides. With the help of trigonometry we can investigate the truth of abstract things.
5. **Analytical Geometry:** This will investigate the various measurements of changing variables. Calculus is the basis of Analysis.
6. **Various Mathematical Topics** : This is dealt with the various usability of mathematics in various mathematical topics.

Every civilize society's basic objectives are to invent new things related to mathematics. Mathematics need is depended on Society's needs. In a complex society, we

need complex mathematical knowledge. Even forest dwellers with the help of the sun, find the measure of time and they use of skills of mathematics for hunting.

In this era of Mathematics, people are astonished to know its use. Branch of Mathematics namely Applied mathematics help solutions for social problems. Mathematics is used to find the solutions for science problems. Mathematics is helpful for the development of Physical Mathematics, Biological Mathematics, Economical Mathematics and Mathematics Technology. Society's development Depends on Mathematics truth and its understanding.

Mathematics therefore is not only computation but it is more about forming generalisations, seeing relationships and developing logical thinking and reasoning.

1.2 Nature of Mathematics

The nature of mathematics is made explicit by discussing it under the following heads.

1. Mathematics - a science of discovery
2. Mathematics - a intellectual game
3. Mathematics - the art of drawing conclusions
4. Mathematics - a tool subject
5. Mathematics - a system of logical processes
6. Mathematics - an intuitive method

1. Mathematics - a science of discovery

Mathematics is the discovery of relationships and the expression of those relationships in symbolic form - in words, in numbers, in letters, by diagrams, or by graphs. According

to A.N. Whitehead, "Every child should experience the joy of discovery". Initially a child's discoveries may be observational. But later when its power of abstraction is developed, it will be able to appreciate the true of the Mathematical conclusions that it has drawn. This will give it the joy of discovering Mathematical truths and concepts.

2. Mathematics - a intellectual game

Mathematics can be treated as an intellectual game with its own rules and without any relation to external criteria. From this view point, Mathematics is mainly a matter of puzzles, paradoxes and problem solving - a sort of healthy mental exercise.

3. Mathematics - The Art of drawing conclusions

One of the important functions of the school is to familiarise children with a mode of thought which helps them in drawing right conclusions and inferences.

4. Mathematics - As a tool subject

From 19th Century, Mathematics achieved a completeness and internal consistency that it has not known before. Mathematics continued to be useful to other disciplines. Its mentors of the past, Engineering, Physics, Science and Commerce now became no more than its peers. Mathematics is a powerful and incisive tool applied in all arts and science subjects.

5. Mathematics - A system of Logical process

Poly suggested that mathematics actually has two faces. One face is "Systematic deductive science" and the another is "inductive science".

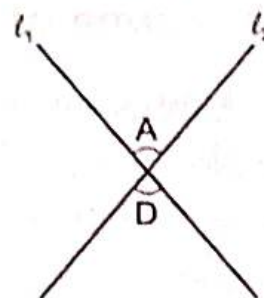
Mathematics is an axiomatic body of definitions, axioms and theorems. Mathematics is a system of logical processes

where by conclusions are deducted from certain fundamental assumptions and definitions that have been hypothesised. The student draws the inferences from the premises, provided the premises are true. For example,

When two lines intersect, vertically opposite angles are equal (the premise)

$\angle A$ and $\angle D$ are vertically opposite angles.

Hence $\angle A$ and $\angle D$ are equal (the inference). This conclusion is deducted from the premise "vertically opposite angles are equal".



The process of deduction involve two steps with logical reasoning. By reasoning we prove that if $a = b$ and $b = c$ then $a = c$. However the validity of the conclusions rests upon the validity and consistency of the assumption and definitions upon which the conclusions are based.

It is based on the principle if a relationship holds good for some particular cases, it holds good for any similar case and hence the relationship can be generalised, such a process is called inductive reasoning. For example, student generalizes that the "sum of the angles in a triangle is 180° ". After having observed this property in a number of triangles, a generalization, a rule or formulae is arrived. Many of the mathematical rules or formulae are generated through induction.

6. Mathematics - An Intuitive method

Intuition implied the act of grasping the meaning or significance or structure of a problem without knowing clear rules or axioms. It is the intuitive mode that yields hypothesis quickly. It precedes proof; it is what the techniques of analysis and proof are designed to test and check. It is a

form of Mathematical activity which depends on the confidence in the applicability of the process rather than upon the importance of right answers all the time. The first steps in the learning of any mathematical subject is the development of intuition.

1.3 Scope of Mathematics

Mathematics provides us inductive and deductive knowledge to enrich our imagination and modes of thinking and even behaviour to excel in life. The material progress of the present day world is the work of our Mathematicians to improve the quality of life by reducing time, distance and human effort. For example computer the new magic of mathematics has a wonderful contribution towards human growth and development.

Almost every subject of study depends directly or indirectly on the application of Mathematics. In Physics, Chemistry, Astronomy, Geology, Engineering etc. Mathematics has vital scope for scientists to design their experiments and analyse the data to its perfection. Social sciences such as Economics, Psychology, Sociology etc., directly depends on the preparation of various models offering strategies, statistical aids, formula so on to make sound decisions for predictions, planning and allied things. It is possible by extending the scope of mathematics according to our hypothesis and designs and approaches to find required solutions to the problems facing day to day activities.

Scope of Mathematics has rapidly increased all the more due to its study as a compulsory subject in the X Std. in India. It is an important subject. In our daily life Mathematics is used largely than other subjects. Educated and uneducated use mathematics in their daily life.

The Education Commission of India state the importance of Mathematics as follows. "We cannot overstress the importance of Mathematics in relation to science, education and research. This has always been so, but at no time has the significance of Mathematics been greater than today. It is important that deliberate effort is made to place India on the 'World map of Mathematics' within next two decades or so".

The World of today depends more and more on Science and Technology. Science and Technology demands more and more mathematical knowledge on the part of more and more people. And the world of tomorrow will still make greater demands on a person with mathematical literacy. Mathematics relies on both logic and creativity and it is pursued both for a variety of practical purposes and for its intrinsic interest. For scientists and engineers, the chief value of mathematics is how it applies to their own work. In modern world some basic understanding of the mathematics is required not only for scientific literacy but also in all fields namely business, agriculture, Industry, administration, medicine and social and Natural sciences.

Mathematics is the basic for our growth in science and technology. Because of mathematics, today we lead a comfortable and happy life. There is a demand for people having good mathematics knowledge for employment in all fields.

1.4 Need and Significance of Teaching Mathematics

Mathematics is the queen of all sciences. The aims and objectives of mathematics teaching has undergone an immense change in the computer era. In the growth and development of an individual and to apply mathematics knowledge and skills in the working place, we need

mathematics. We can not give the importance of mathematics in a few lines. Any how the following points will tell us the need and importance of mathematics.

- To know the world around us
- To enable the individual to meet the demands of daily life.
- To develop skill in the use of mathematical tables, statistical tables and computers.
- To understand the culture, social development and civilization.
- To understand the natural phenomena clearly
- To develop self confidence, self reliance & patience
- To apply mathematical concepts and principle in a new situations.
- To apply mathematics knowledge and skills in the working place.
- To apply mathematics knowledge in the other school subjects.
- To enable the pupils to appreciate the subject of mathematics in Engineering and technology.
- To find the alternate methods while solving a problem
- To develop the skill to express mathematical thoughts clearly and accurately.

1.5 Values of Teaching Mathematics

To teach any subject, it is the primary fact to know the teachers and students, what is the purpose of teaching and what are the values of teaching. Values give meaning and

strength to a person's character. Values are guiding principles of life which are conducive to all round development. Therefore values reflect once personal attitudes, judgements, decisions, choices, behaviour, relationships etc. Napoleon remarked that the progress and improvements of Mathematics is linked to the prosperity of the state. Therefore Mathematics plays an important role in the progress of society. A few values of Mathematics teaching is given below :

1. Intellectual values
2. Practical values
3. Social values
4. Disciplinary values
5. Cultural values
6. Moral values

1. Intellectual value

Mathematics teaching is very important for intellectual development. There is no other subject in the curriculum like Mathematics which make students' brain active. Problem - solving helps us in development of mental faculties. Mental work is needed for solving mathematical problems. Brain becomes active in solving Mathematical problems. Each problem of mathematics possesses such a sequence which is necessary for constructive and creative process. In this way all mental abilities of child are developed through mathematics. Plato has said, "Mathematics is the subject which provides an opportunity for training in mind."

World is filled of broad knowledge which is increasing day by day. It is not so important to achieve knowledge, but to learn how to achieve knowledge is more important knowledge depends upon mental abilities of students. It

provides all the opportunity to the students to develop their observation power, logical power, memory, concentration, originality, power of discovery, thinking power, reasoning power, self reliance and hard work etc.

2. Practical values

Practical values deals the relationship between members and day to day activities of our life. Next to mother tongue mathematics is related in all our activities of life. A man can live without education but not without mathematics. From the daily labourer to the Proprietor Mathematics is the success for their development. Without mathematics there is no life in this world. We need its knowledge in our daily routine numerical problems, percentage, profit & loss etc. For example to purchase a land, to sell a land or to divide a land one must know the computational knowledge, profit and loss, ratio etc of mathematics. To study other school subjects especially science subjects we need the knowledge of mathematics. News paper items such as economical growth, industrial growth, sensex index, Dearness allowance etc. can be known only if we have mathematics knowledge. To understand the statistical graphs which appears in Newspaper, we need mathematics knowledge. All scientific instruments which have made our daily life so easy smooth and happy are only by mathematics. If students realize how mathematics is used in our daily life, then they will study mathematics with interest.

3. Social Values

Social values deals with how mathematics is used in maintaining and administering general social and economic life of his country. To administer social industries like Central and State Government administration, cooperative societies, Banks, Insurance companies, industries etc. mathematics knowledge is needed to make decisions in human affairs and

official affairs. Mathematical methods such as scientific method, laboratory method, logical method, deductive method etc. are used to analyse, interpret social values. Statistics and probability are used to deal the above concepts. Napoleon said "The progress and improvements of mathematics are linked to the prosperity of the state".

Almost all the departments like business, banks, Engineering, Medicine, Tax, Insurance, Administration totally depend on Mathematics. Those who do mistakes in mathematics will not succeed in their business. The contribution of Mathematics is great in the modern society. Through the study of mathematics, we get self confidence, satisfaction, self reliance, confidence, patience, strong determination, to do work systematically, helping others, leadership qualities etc all this leads to live a satisfactory life in the society. The self confidence which we get through the study of mathematics has more value. Mathematics teaches the art of economical living.

4. Disciplinary Values

Disciplinary values deals with how mathematics is used to do any work with self confidence and discipline. The two main objectives of learning mathematics are i) to get knowledge ii) To learn disciplinary methods. Lockey says, "Mathematics is a way to settle in the minds of children, a habit of reasoning". So learning mathematics is for reasoning power and to lead a systematic life. For a discipline life we need the qualities like simpleness, logical thinking, intelligent, hardwork, patience, concentration, well organized etc. Learning mathematics will help to develop the above qualities. We teach mensuration to relate the steps, to determine a solution, to emphasise exactness, to attain a sense of appreciation etc. We teach algebra to compile general laws and think abstract concepts etc. We teach geometry to deduct correct results from logical thinking.

While solving a problem a student gets the above disciplined values.

Instead of knowing mathematical knowledge, a person who uses mathematical methods matters most. You may forget the methods of factorisation or Geometrical proof of $(a+b)^2$ etc. But you can't forget the methods of thinking, correlating steps, to verify conclusion etc. These qualities help us to lead a disciplined life and find a better solution of our day to day problems in a disciplinary way.

5. Cultural Values

Progress in Mathematics of Greeks and Egyptians in the past led to their cultural advancement and progress of their civilization. Cultural values deals with how mathematics is used in the development of cultural and civilizational values of a nation. No one can refute the importance of mathematics in the history of world civilization. Educational objectives vary according to the need of the society from time to time. In the beginning education was based on the strength of the body and now it has changed to its depth of knowledge and discipline. Pythagoras said, Physical exercises, music and mathematics are the three stages in educational curriculum. First with the help of physical exercises student become strong. Second people purify themselves with the help of music and third with the help of mathematics they attain perfection. From the above statements, we know how Pythagoras considered mathematics. Hogben remarked that "Mathematics is the mirror of civilization". Science and Technology develops on the development of mathematics and society uses its benefits. Societies culture depend upon their way of life. Way of life depends upon the development of science and technology. So Mathematics is the mirror of civilization. Today we are well developed in the departments of Engineering, Medicine, Industries, Transport, Agriculture,

Atomic energy, Astronomy etc. and it is the contribution of mathematics. Without the higher knowledge of mathematics no one can show progress in science. Hence mathematics plays an important role in the development of culture and civilization of the society.

6. Moral Values

Moral values are the standards of good and evil which governs an individual's behaviour. Science has more moral values. Science is the search for truth in a faithful manner. Promotion of science along with the growth of moral values is necessary for human development. Scientists strongly abhor fraud, error and pseudoscience while they value reliability, testability, accuracy, precision and simplicity. These values are generally derived from our experience in research. Good views of others are accepted by scientists. For the welfare of mankind scientists conduct research and invent new things. It teaches the pupil to be intellectually honest and truthful in our daily activities.

1.6 Aims and Objectives of Teaching Mathematics

AIMS

To teach any subject, it is the primary fact to know the teachers and students, what is the purpose of teaching and what are the uses of teaching. We use the term 'aims' to mean the broad goals which our educational system (the whole of the Indian society involved) embraces and which we are expected to attain. The aims of education are based on philosophical and socio-psychological aspects of society and culture. Aims are long term goals.

Objectives

Objectives are the milestones to reach the destination i.e. to attain the aim or ultimate goal of education. Objectives are specific, direct and practical in nature; objectives are related to the learning outcomes or change in the behaviour of the students. The objectives are meant for both the teacher and the students. Even at the planning stage i.e. before entering the classroom the teacher asks himself, "what changes in behaviour of the students can I bring about through this lesson?" such changes constitute his (teacher's) educational objectives. Objectives are short term goals.

To teach any subject, it is the primary fact to know the teachers and students, what is the purpose of teaching and what are the need and uses of teaching. Mathematics is the queen of all sciences. The aims and objectives of mathematics teaching has undergone an immense change in the computer era. In the growth and development of an individual and to apply mathematics knowledge and skills in the working place, we need mathematics. We can not give the importance of mathematics in a few lines. Any how the following points will tell us the need and importance of mathematics and the aims and objectives of mathematics.

i. Knowledge objectives

1. To know the world around us
2. To understand the natural phenomena clearly
3. To enable the individual to meet the demands of daily life.
4. To understand the culture, social development and civilization.
5. Perform Mathematical operations and manipulations with confidence speed and accuracy.

6. To collect, organize, analyze and interpret data and make conclusions and predictions from its results.

7. To prepare students to college level

ii. Skill Objectives

1. To develop skill in the use of mathematical tables, statistical tables and computers.
2. To apply mathematical concepts and principle in a new situations.
3. To apply mathematics knowledge and skills in the working place.
4. To apply mathematics knowledge in the other school subjects.
5. To develop the skill to express mathematical thoughts clearly and accurately.
6. To Develop Mathematical curiosity and use inductive and deductive reasoning when solving problems.
7. To recognize patterns
8. The real aim of teaching mathematics is to develop power, not knowledge. The knowledge acquired through reasoning becomes power, which helps in discovering new facts.
9. To develop logical and critical thinking
10. To develop various skills like accuracy, neatness in drawing geometrical figures.

iii. Appreciation Objectives

1. To recognize that mathematics is a powerful and fundamental tool of science with limitless applications.
2. To enable the pupils to appreciate the subject of mathematics in Engineering and technology.

3. To enjoy mathematics and develop patience and persistence when solving problems.
4. To exhibit how mathematics is related with other school subjects.
5. To appreciate the culture values of Mathematics

iv. Attitude Objectives

1. To develop self confidence, self reliance & patience
2. Develop a positive attitude towards learning Mathematics.
3. Think precisely, logically and critically in any situation.
4. To inculcate the habit of hardwork, patience, self confidence, regularity etc.
5. To develop moral values of the pupil

v. Social Objectives

1. All round development of the student for improvement of his personality
2. To develop willingness to work collaborately.
3. To develop the leadership qualities.
4. To develop the habit of helping others.
5. To develop self confidence, self reliance & patience.

1.6.1 Aims and objectives of teaching Mathematics in Schools.

It can be divided into three stages

1. Aims and objectives of teaching Mathematics at Primary stage.

2. Aims and objectives of teaching Mathematics at High School stage.

3. Aims and objectives of teaching Mathematics at Higher Secondary stage.

1. Aims and objectives of teaching Mathematics at Primary stage.

- Acquire understanding of numbers and numeration
- Develop ability to perform the four basic operations
- Develop skills in measurement, approximation and estimation.
- Acquire the following knowledge concepts and ability to use them in daily life.
 - ✓ Numbers and basic functions
 - ✓ Number system
 - ✓ Basic concepts of Geometric figures - two dimension and three dimension.
 - ✓ To know the measurement of length, weight, volume and time.
 - ✓ To know our currency.
- Acquire the techniques of collecting representing and interpreting data.
- Develop the habit of systematic way of thinking
- To relate the fundamental mathematical knowledge and skills to daily life situation.
- Develop positive attitude towards mathematics and make good use of leisure time.
- Develop techniques of investigation and problem solving

- To develop the habit of speed, accuracy and using mathematical symbols.

2. Aims and objectives of teaching Mathematics at High School stage.

- To develop mathematical curiosity and use inductive and deductive reasoning when solving problems.
- Understanding the concepts from the five branches of mathematics (Number system, Algebra, Geometry and Trigonometry, Statistics and probability and discrete mathematics).
- Develop abstract, logical and critical thinking and the ability to reflect critically upon their work and the work of others.
- Use appropriate mathematical concepts and skills to solve problems in both familiar and unfamiliar situations including those in real life situations.
- Select and apply concepts and rules correctly to solve problems
- Appropriate use of Mathematical symbols in both oral and written explanations.
- Develop the knowledge, skills and attitudes necessary to pursue for the studies in Mathematics.
- Suggest improvements to the method when necessary
- Collect organise represent, analysis and interpret data and make conclusions and predictions from its results.
- To develop habits and attitudes like habit of hard work, self reliance, concentration and discovery.
- Use technology to enhance mathematical thinking

- Develop willingness to work collaborately

3. Aims and objectives of teaching Mathematics at Higher Secondary stage.

Mathematics is taught as a separate subject in higher secondary stage.

- i. To familiarize the pupil with the world in which he lives and to make him understand the impact of Mathematics on society.
 - ii. To develop the scientific attitude
- Followers to cause and effect relationship
 - Adopts planned procedure in solving a problem
- iii. Fostering of creativity leading to innovations in Mathematics .
 - iv. To acquaint him with the 'scientific method' and to enable him to develop the scientific attitude.
 - v. Developing interest in Mathematics hobbies. Appreciate the patterns of Mathematics in leisure time.
 - vi. Understanding the knowledge and concepts of complex numbers, vector analysis, analytical geometry, calculus, probability and their applications.
 - vii. Develop the knowledge skills and attitudes necessary to pursue further studies in mathematics.
 - viii. Make conclusions and predictions from the results.
 - ix. Develop willingness to work collaborately.
 - x. To appreciate the Culture and Values of Mathematics.

1.6.2 Difference between Aims and Objectives

Sl. No.	Aims	Objectives
1.	Aims are long term goals	Objectives are short term goals.
2.	Aims are broad and common	Objectives are narrow and specific
3.	Philosophy, sociology is main source of aim	Psychology is the main source of objectives
4.	They are difficult to achieve	They can be achieved conveniently
5.	These can't be evaluated	These can be evaluated
6.	School, Society and Nation is responsible to achieve them	Generally teacher is only responsible
7.	Aims are non observable behaviours	Objectives are observable behaviours.
8.	To achieve aims we have to go beyond curriculum	Objectives are achievable within the curriculum.

1.7 Instructional Objectives and Behavioural Objectives

Instructional Objectives

Instructional objectives are basically statement which clearly describe an anticipated learning outcome.

Instructional objectives are helpful to the teacher as well as the learner throughout the learning process and are invaluable in the evaluation process. An instructional objective is the focal point of a lesson plan. Before start teaching teacher must be guided by instructional objectives. Instructional objectives can be classified into three domains of learning i.e. cognitive, affective and psychomotor. Objectives should state what the learners should be able to do after the instruction is complete. The process of how the instruction happens is not considered as an objective.

Purpose of Writing Instructional Objectives

- Define the desired outcome of instruction in terms of tasks that student will be able to perform.
- Establish expectation for the student.
- Focus course development and teaching activities on relevant information and skills.
- Provide a clear reason for teaching
- Provide criteria for eliminating unnecessary information and activities
- Indicate how success can be measured

Characteristics of Instructional objectives

Instructional objectives are specific, measurable, short term, observable student behavior. They indicate the desirable knowledge, skills of attitudes to be gained at the end of teaching. It describe the performance of a learner should be able to demonstrate after instruction.

- A well-written objective should describe a learning outcome. It should not describe a learning activity
- An objective focuses on the learner, not on the teacher. It describes what the learner will be expected to be able to do. It should not describe a teacher activity.
- If an instructional objective is not observable, it leads to unclear expectations and it will be difficult to determine. The key to writing observable objectives is to use verbs that are observable and lead to a well defined product. Verbs such as "to know" to understand, "to enjoy" to appreciate etc. are vague and not observable. Verbs such as "to identify", "to

list", "to compute" are explicit and describe observable actions or actions that lead to observable products.

- There are many skills that cannot be directly observed. The thinking process of a student to solve a mathematics problem cannot be easily observed. However one can look at the answers or the steps to arrive at an answer.
- Objective to be sequentially appropriate
- Objective to be attainable within reasonable amount of time.
- Objective to be developmentally appropriate (within students level of intellectual)

Importance of Instructional Objectives

Instructional objectives are important to both learners and instructors. They help learners plan their study and prepare for examinations. They guide the instructors in planning instruction and devising tests. Instructional objectives play an important role in the teaching and learning process. They are the heart of teaching.

Importance of Instructional Objectives

1. To provide direction for the instructional process by clearly stating the intended learning outcomes.
2. To convey instructional intent to students, parents and educational authorities.
3. To provide a basis for evaluating student's learning by describing the performance to be measured.
4. They state in clear terms behavioural changes expected from a learner after with a learning activity.
5. Without instructional objectives, it will be very difficult for a teacher to access the behaviours of the students in the

three domains of education. (Cognitive, affective and Psychomotor Domains)

6. Instructional objectives encourage reflection and good course design and development.
7. Instructional objectives help to define reinforcement situations of the learners.
8. They enable the teacher to modify methods to suit the process of eliciting those behaviours that the objectives specify.
9. They help the teacher to know how well the students have covered the content.
10. They give direction to both teacher and students in the selection and use of materials, methods and activities for learning.

Behavioral objectives

A behavioral objective is a learning outcome stated in measurable terms, which gives direction to the learner's experience and becomes the basis for student evaluation.

Points in writing behavioral objectives

1. Begin each behavioural objectives with a verb. The critical aspect of any behavior is the verb selected from learning activities.
2. State each objective in terms of learner performance. A behavioral objective is one that is considered to be observable and measurable. Behaviour is generally constructed to be an action of an individual that can be seen, felt or heard by another person.
3. State each objective so that it includes only one general learning outcome.

Examples of Behavioral Objectives

- Knowledge : Remembering previously learned facts
- Comprehension : Ability to understand or grasp the meaning of material.
- Application : Ability to use previously learned material in new and concrete situations.

Motivation and Behavioural Learning

Motivation plays an important role in behavioural learning. Positive reinforcement can be motivators for students. For example, a student who receive praise for a good test score is much more likely to learn the answers effectively than a student who receives no praise for a good test score. The student who receives no praise is experiencing negative reinforcement - their brains tell them that though they got a good grade, it did not really matter, so the material of the test becomes unimportant to them. Conversely students who receive positive reinforcement see a direct correlation based on that response to a positive stimulus.

Purpose of Behavioural objectives

- Guide for the teacher to design of instruction.
- Guide for the teacher for evaluation
- Guide for the learner relative to learning focus
- Guide for the learner relative to self assessment
- Makes teaching more directed and organized
- Forces teacher to think carefully about what is important.

- Helps teacher to avoid unnecessary repetitions in teaching.
- Provides visibility and accountability of decisions made by teachers and learners
- Helps students make decisions regarding prioritizing
- Helps relationship between teacher and learner.

Examples of Behavioural objectives

General Objectives : The learner will be able to Solve Quadratic Equation

Specific Objectives : The learner will be able to solve Quadratic Equation using formulae method.

1.7.1 Writing Instructional Objectives in terms of Behavioural Objectives

A behavioral objective is a way by describing the objectives of a content in terms of what the learner should be able to do at the end of the lesson. A behavioural objective must be stated clearly and precisely so that everyone who reads it will know exactly the desired outcome of the content.

In writing a behavioural objective, it is important to include the verb that depicts what the learner is expected to do, i.e. it should be measurable.

Examples

- 1) The students will be able to describe the process of transpiration in plants

Here the students are going to describe at the end of instruction. Here the emphasis is on learners and the learning outcome. Here learning is a change of behavior.

- 2) To identify proper fractions

Here the learner and his performance are given.

- 3) The students will understand the three laws of reflection

Here the verb understand does not indicate behavior that can be observed directly. It does not indicate any specific behavior. If the objective is written as.

The students will tell the three laws of reflection.

Then it is observable behavior. Hence the verb 'understand' can further be explained and classified by listing under that some specific behavior like tell, identify, describe, give, differentiate etc. In other words the verb understand is more general and the verb tell is more specific as compared to other.

Let us discuss the two objective in details

- i. The pupil will know the working of a steam engine.
- ii. The pupil will be able to describe in about ten to fifteen sentences the working of a steam engine.

In the above example i) one cannot see other's mind to determine what he knows. We are not able to observe some aspect of this behavior that corresponds his knowledge. Hence an objective should indicate the kind of performance which will be accept as evidence.

Thus in the second example, well-stated instructional objectives are given in observable behavioral terms.

Few examples of well-stated Instructional Objectives

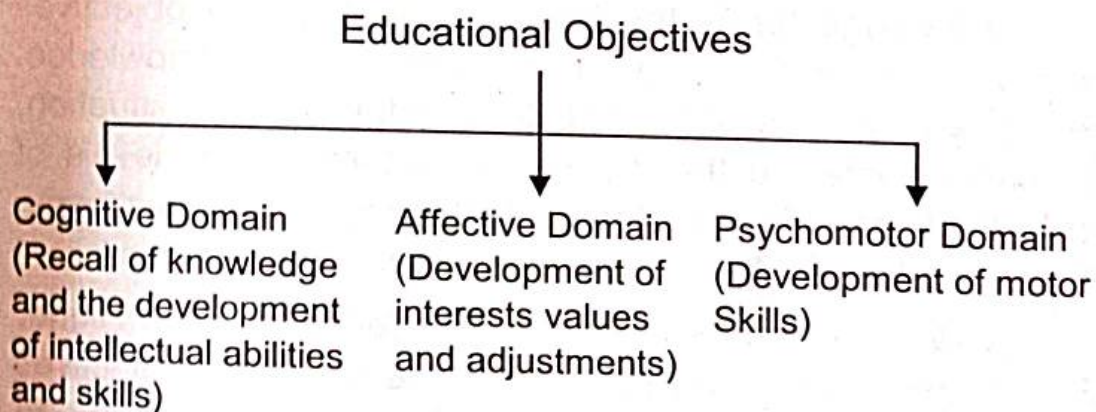
1. The students will be able to describe the three causes of World War II as discussed in the class.
2. The class will be able to identify correctly, all the parts of the hibiscus flower.
3. Students will tell the name of any four corporations in Tamil Nadu.

4. Students will be able to identify the part of a simple sentence.
5. Students draw a right angled triangle with given two sides.

1.8 Blooms Taxonomy of Educational Objectives

To define understand and classify the educational objectives American educationalist Benjamin S. Bloom and his associates at the University of Chicago divide it into three domains i.e. cognitive, affective and psycho-motor and explains clearly to the teachers.

A brief account about these is given below.

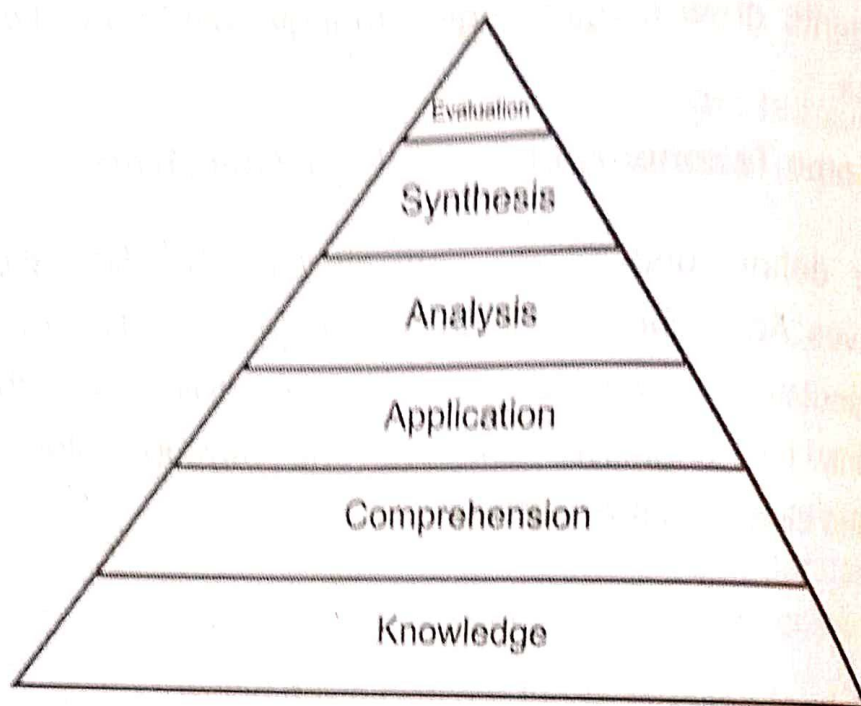


In each domain, objectives are hierarchical order starting from easy to difficult. Hence this arrangement is called "Taxonomy of Educational objectives.

1.8.1 Cognitive Domain

Cognitive domain includes those objectives which deal the recall or recognition of knowledge and the development of intellectual abilities and skills. This domain consist of six

parts and arranged in a hierarchical order, in increasing levels of difficulty (from lower to upper)



Knowledge forms the basis for all cognitive objectives and all other objectives are not possible without knowledge. Similarly if one is able to apply knowledge in a new situation, it can be inferred that he has to achieve lower levels of objectives namely knowledge and comprehension clearly. Thus a higher level objective subsumes to lower ones.

While teaching mathematics for Secondary and Higher Secondary students give priority to the first three objectives of cognitive domain namely knowledge, comprehension and application. The remaining three objectives Analysis, Synthesis and Evaluation may be trained to them according to their maturity of mind.

Knowledge :

This objective is to remember and recall or recognize the already learnt information and acquire more and more knowledge.

Comprehension :

Objective at this level requires learners to understand the knowledge without any doubt. He should have ability to arrange data in order.

Application :

This objectives requires students to apply already learnt knowledge to a new and unfamiliar situations. Think critically in depth.

Analysis :

Objective at this level requires the learners, to attain the ability to break up a given communication into parts to understand it better.

Synthesis :

Synthesis is just opposite to Analysis. The ability to draw upon elements from various sources and put them together to produce a new structure. This objective gives divergent thinking and creativity.

Evaluation :

This is a highest objective of cognitive domain. The ability to make judgements about something for given purpose. Judgement can be both quantitative or qualitative.

It helps teachers in the following ways:

- Helps to assess and judge the difficulty of assignments.
- It increase the accuracy of the lesson
- Simplifies action to ease personal learning
- Designed a combined assessment
- Helps for project based plans for learning
- Helps for group discussion

Instructional objects of Cognitive Domain

Parts of Cognitive Domain	Instructional Objectives
<u>Knowledge</u> Student knows the mathematical symbols, concepts, definitions, formulas, statements etc. of mathematics subject	<u>Student</u> 1. Recalls the mathematical formulas, concepts and statements. 2. Recognises what he learned.
<u>Comprehension</u> Student understand the mathematical symbols, concepts, definitions, formulas, statements	<u>Student</u> 1. Gives examples 2. Give reasons 3. Compares 4. Differentiates 5. classifies 6. relates 7. detect errors and rectify 8. using symbols 9. Explains 10. Evaluate the result
<u>Application</u> Using mathematics knowledge and comprehension to solve the problem in a new situation.	<u>Student</u> 1. Examine what is given in the problem? What is need? 2. Examine the sufficiency of the data. 3. Select appropriate formula 4. Solve the problem in an appropriate method 5. Applying alternate method 6. Verifying the result.
<u>Analysis</u> Break up a given mathematics statement into small meaningful parts	<u>Student</u> 1. Understand the meaning of the hypothesis 2. From the given data draw conclusion

Parts of Cognitive Domain	Instructional Objectives
	- Identify motives / causes for the mathematical concepts. - Discriminates, selects, illustrate mathematical concepts.
<u>Synthesis</u> Putting elements from various sources and put them together to produce a new structure	<u>Student</u> - Design a method to perform a new task. - Revises the process to improve the outcome. - Compile, generate, create mathematical concepts. - Solves the problem - Develop the idea in advance.
<u>Evaluation</u> Evaluating Qualitative or quantitative judgement for mathematics elements or methods.	<u>Student</u> - Evaluate the solution of a given problem - Select the most effective solution. - Explain and justify a new proposal - Assess / defend mathematical concepts. - Review / Criticise mathematical statements.

1.8.2 Affective Domain

The affective domain refers to emotional and attitudinal engagement with the subject matter while the cognitive domain refers to knowledge and intellectual skills related to the subject matter.

Affective domain includes the development of feelings of the educational and social attitudes and to increase the emotional objectives namely attitude, interest, values and appreciation. The different objectives under affective domain are Receiving, Responding, Valuing, organisation and characterisation.

1. Receiving:

This is the first step for the learner to learn what the teacher teaches. Listen attentively to a particular phenomena or stimuli.

2. Responding:

This is also called interest objective. Pupil is sufficiently motivated to attend actively. Interest in mathematics lends the student to select a particular vocation throughout his life. Interest can be developed in solving puzzles, reading mathematical books and journals etc.

3. Appreciation :

Student appreciates Mathematics for its use in daily life and how it helps to other subjects.

4. Organisation :

This objective brings together different values, resolving conflicts among them, starting to build an internally consistent value system.

5. Valuing:

It means individual's own valuing or assessment of thing, phenomena or behaviour. It is also called attitude. Development of proper attitude in mathematics is one of the major objectives of teaching mathematics.

Instructional Objectives of Affective Domain

Parts of Affective Domain	Instructional Objectives
<u>Receiving</u> Listen to others with respect. Listen for and remember the newly introduced content.	<u>Student</u> 1. Identifies 2. Asks 3. Understand 4. Points out errors boldly. 5. Listen to others attentively.
<u>Responding</u> Participate in class discussions. Gives a presentation, questions new ideas, concepts etc.	<u>Student</u> 1. Reads books of mathematics 2. Solves mathematics puzzles 3. Writes activities on mathematics for school magazine. 4. Gives short cuts for solving problems. 5. Does problems apart from syllabus. 6. Gains mathematics knowledge more than the syllabus. 7. Participates in mathematics club activities. 8. Participate in class discussions.
<u>Appreciation</u> Valuing to a particular object, phenomenon or behavior ranges from simple acceptance to the more complex state of commitment.	<u>Student</u> 1. Solving the problems of other subjects using mathematics. 2. Appreciates the symmetrical pattern of figures seen in his daily life. 3. Appreciates how mathematics is useful in his daily life.
<u>Organisation</u> Recognise the need for balance between freedom and	<u>Student</u> 1. Recognize the need for balance between freedom and responsible behavior

Parts of Affective Domain	Instructional Objectives
responsible behavior. Explains the role of systematic planning in solving problems.	2. Explain the role of systematic planning in solving the problems. 3. It becomes habit to think logically.
<u>Valuing</u> Has a value system that controls his behavior. Uses an objective approach in problem solving. Revises judgments and changes behavior in the light of new evidence.	<u>Student</u> 1. Shows self reliance when works independently 2. Co-operates in group activities. 3. Revises judgements and changes behavior in light of new evidence. 4. Values people for what they are not how they look. 5. Accepted statements when logically proved.

1.8.3 Psychomotor Domain

Elizebath Simpson built her taxonomy on the work of Bloom and others. Simpson's Psychomotor domain is comprised of utilizing motor skills and coordinating them. Simpson's taxonomy has a focus toward the progression of mastery of a skill from observation to invention.

The psychomotor domain includes Physical movement, coordination and use of the motor-skill areas. Development of these skills requires practice and is measured in terms of speed, precision, procedures or techniques in execution. The seven major categories are perception, set, Guided Responses mechanism, complex over to Response, Adaptation and Origination.

The behaviour and development of every child is depends on the above three domains. So while teaching, the teachers teach the subject matter on the basis of the above three domains.

1. Perception

It means observing the act of the teacher.

2. Set

Readiness to act. It includes mental, physical and emotional sets

3. Guided Response

Following directions of the teacher and differentiating among various movements. This is the early stage in learning complex skill.

4. Mechanism

This is the intermediate step in learning a complex skill. Learned responses have become habitual and the movements can be performed with some confidence and proficiency.

5. Complex Overt Response

This is the highest level of proficiency and it becomes natural and automatic. Proficiency is indicated by a quick, accurate and highly coordinated performance, requiring a minimum energy.

6. Adaptation

Skills are well developed and the individual can modify movement patterns to fit special requirements

7. Origination

Creating new movement patterns to fit a particular situation or specific problem. Learning outcomes emphasize creativity based upon highly developed skills.

Instructional Objectives of Psychomotor Domain

Parts of Psychomotor Domain	Instructional Objectives
Perception Perception is the most basic level to process sensory information that guide motor activity	Student 1. Detects non verbal communication cues 2. Estimate where a ball will land after it is thrown and then moving to the correct location to catch the ball. 3. Observing the act of the teacher.
Set Mentally, emotionally and physically ready to act.	Student 1. Repeat the act of the teacher. 2. Acts upon a sequence of steps to solve a problem. 3. Shows desire to learn a new process.
Guided Response Imitates and practices skills often in discrete steps	Student 1. follows 2. react 3. response 4. copies 5. performs
Mechanism Performs acts with increasing efficiency, co-ordinate and proficiency.	Student 1. Drives a cycle 2. assembles 3. dismantles 4. measures 5. mixes
Complex Overt Response Motor activity performs automatically.	Student 1. Operates a Computer quickly and accurately

Parts of Psychomotor Domain	Instructional Objectives
	2. assembles 3. builds 4. measures 5. organizes
Adaptation Adapts skill sets to meet a problem situation.	Student 1. Responds effectively to unexpected situations. 2. Modifies method to meet the needs. 3. Perform a task with a new method.
Origination Create new patterns for specific situations.	Student 1. Constructs a new theory. 2. designs 3. creates 4. builds 5. composes

Skills to be developed in mathematics

In mathematics the following skills may be developed.

Skill of drawing figures

Skill of drawing graphs

Skill of reading tables

Skill of computation

Parts of Psychomotor Domain	Instructional Objectives
Student develops skill in drawing geometrical figures.	Student 1. Draws rough figure with given data. 2. selects appropriate geometrical instruments

Parts of Psychomotor Domain	Instructional Objectives
	- draws geometrical figures with given measurements. - draws neatly and accurately - measures with speed and accuracy.
Student develops skill in drawing graphs	<u>Student</u> - selects appropriate scale - plot the points neatly and accurately - draws the graph neatly and accurately - Interprets the graph correctly.
Student develops skill in reading table	<u>Student</u> - reads the mathematical tables correctly - reads to table with speed and accuracy
Student develops skill in computation	<u>Student</u> - does oral sums with speed and accuracy - does written calculations with speed and accuracy - operates a computer quickly and accurately - use scientific tables quickly and accurately - does written calculations neatly.

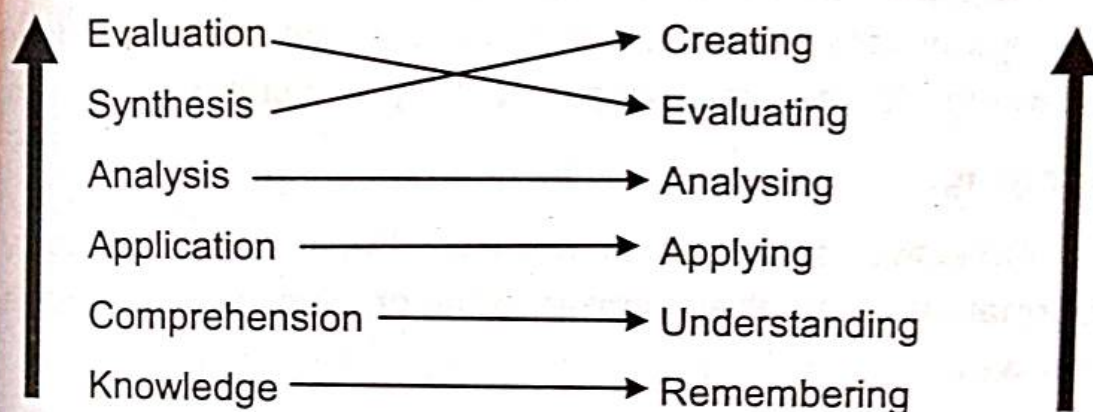
The behaviour and development of every child is depends on the above three domains. So while teaching, the teachers teach the subject matter on the basis of the above three domains.

1.9 Revised Bloom's Taxonomy

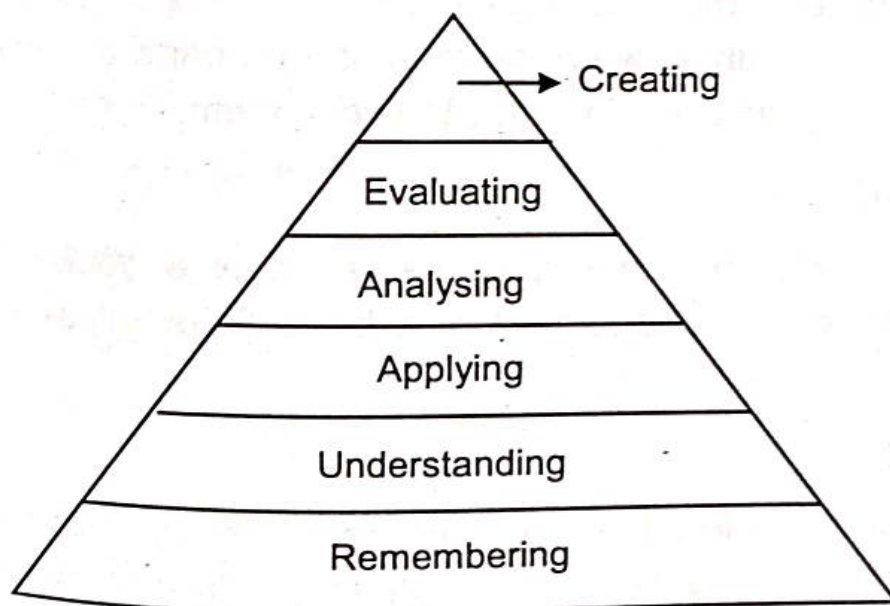
In 1990 Loris Anderson former student of Bloom revised the taxonomy and made a number of changes.

Bloom's Taxonomy

Revised Bloom's Taxonomy



Revised Bloom's Taxonomy



Remembering:

The knowledge category was renamed as remembering. Knowledge is an outcome or product of thinking and not a

form of thinking. Hence the word knowledge was inappropriate to describe a category of thinking and was replaced with the word remembering.

Remembering is recalling information. This objective is to recognizing, listing, describing, naming, finding already learnt information.

Understanding:

Objective at this level requires learners to understand the knowledge without any doubt. He should have knowledge to interpret, explain, classify and summarize.

Applying:

Objective at this level requires the learners to apply information in another familiar situation and executing in a best way.

Analysing:

Objective at this level requires the learners to attain the ability to break up a given information into parts to explore understanding and relationships between them.

Evaluating:

The ability to make judgements about a decision or course of action. Judgement can be both quantitative or qualitative.

Creating:

By putting elements together from various sources to form the ability to generate new ideas, products or ways of viewing thing in a different way. Designing, planning, producing, inventing new things and objects is the objective at this level.

Potential activities of cognitive domain**Remembering:**

1. Make a list of the main events of today.
2. Write any ten formulae in menstruation.

Understanding:

1. Retell the story in your own words.
2. Prepare a flow chart to illustrate the sequence of events.

Applying:

1. Construct a model to demonstrate how it works.
2. Make a clay model to explain a mathematical concept.

Analyzing:

1. Design a questionnaire to gather information.
2. Construct a graph to illustrate given information.

Evaluating:

1. Conduct a debate about an issue of
2. Prepare a case to present your view about.....

Creating:

1. Invent a machine to do a specific task.
2. Design a magazine cover for sports day function.

1.10 Writing instructional objectives for different mathematics Topics

To write objectives for a particular mathematics topic, begin with a action verb while writing objectives. Do not use verbs such as learn, understand, Discuss, explain etc, as these do not specify exactly how students will demonstrate

their knowledge. use specific action verbs that will demonstrate exactly what the student will be doing during the lesson.

Remember that your objectives must be observable. Can you see a student determine? Can you see a student understand? Do not use unobservable behaviors as objectives.

1. Std : 8

Topic : Congruent triangles

Instructional Objectives of Teaching

1. Knows the conditions (knowledge) for the two triangles to be congruent.
2. tells the conditions for the two triangle to be congruent.
3. Using knowledge and understanding solves a problem in a new situation
4. Learns the skill to cut congruent triangles in paper.
5. Approaches the congruent triangles with a scientific attitude.
6. Solves the problem with interest.
7. Applies the skill to verify two triangles to be congruent.
8. Writes oral informations into symbols.
9. Appreciates the congruent pattern.
10. Cuts right angles and equilateral triangles in colour papers for Annual day of the school.

2. Std : 10

Topic : Volume of the Cone

Instructional Objectives of Teaching

1. Knows the method of finding volume of the cone.

2. tells the method of finding volume of the cone.
3. Using knowledge and understanding solves a problem in a new situation.
4. Obtains the skill to find volume of the cone shaped things.
5. Approaches the various aspects of the problem with a scientific attitude.
6. Solves the problem with interest.
7. Finds the relationship between the volumes of cylinder and cone having same radius and height.
8. Gives examples of cone shaped things.
9. Does written calculations with speed and accuracy
10. Finds volume of the cone shaped watertank.

3. Std : 8

Topic : Drawing rhombus

Instructional Objectives

1. Knows the method of drawing rhombus.
2. tells the method of drawing rhombus.
3. Using knowledge and understanding solves a problem in a new situation.
4. Obtains the skill of draw rhombus type figures.
5. Draws the rhombus with a scientific approach
6. Draws the rhombus with interest
7. Draws the rhombus accurately in order

8. Uses appropriate geometrical instruments
9. Analyses what is given? and what is asked? in a problem
10. Finds area of the rhombus accurately

4. Std : 8 Topic : Real numbers - Four properties of Addition and subtraction of rational numbers

Instructional Objectives

1. Knows four properties of addition and subtraction of rational numbers.
2. tells the four properties of addition and subtraction of rational numbers.
3. Using knowledge and understanding solves a problem in a new situation.
4. Analysis the closure property, associative property in respect of addition and subtraction of rational numbers
5. Improves computation skill
6. Solves the problem with interest
7. Differentiates between additive identity and subtractive identity.
8. Obtains skill to tell additive inverse of any rational number
9. Analysis the various components of a given problem
10. Solves the problem in an appropriate way and get the result.

5. Std : 9

Topic : Algebra - Factorization

Instructional Objectives

1. Knows the method of factorization.

2. tells the method of factorization.
3. Using knowledge and understanding solves a problem in a new situation.
4. Obtains the skill to factorize algebraic equation
5. Analysis various methods of factorization with a scientific approach.
6. Solves the given problem with interest.
7. Compares the terms of the given polynomial.
8. Analyses the problem and solve using appropriate method.
9. Analyses all the aspects of the problem.
10. Solves the problem in an appropriate method and gets the result.

1.11 Inter-relation among the Domains

Learning helps develop an individual's attitude as well as encourage the acquisition of new knowledge and new skills.

Learning can generally be categorized into three domains: Cognitive, affective and psychomotor. Learning is not an event. It is a process. It is the continual growth and change in the brain's architecture that results from the many ways. We take in information, process it, connect it, catalogue it and use it.

Cognitive Development

Cognitive development refers to the way a child grows intellectually. Assessing for academic skills, reasoning, and critical thinking abilities, and the acquisition of knowledge or learning. Difficulties interacting with others as a result of problems and social or emotional development will impede a

child's ability to focus. Problems or delays in physical development might impact a child cognitively.

Affective Domain

The affective domain includes the feelings, emotions and attitudes of the individuals. The categories of affective domain include receiving phenomena; responding to phenomena; valuing; organization and characterization. The sub domain of receiving phenomena creates the awareness of feelings and emotions as well as the ability to utilize selected attention. This can include listening attentively to lessons in class. The next sub domain of responding to phenomena involves active participation of the learner in class or during group discussion. The ability of the student to prioritize a value over another and create a unique value system is known as organization. This can be assessed with the need to value one's academic work as against their social relationships. This helps to control the behavior of the individual.

Psychomotor domain

This includes utilizing motor skills and the ability to coordinate them. The sub domains are perception; set; guided response; mechanism; complex overt response; adaptation; and origination. Perception involves the ability to apply sensory information to motor activity. for eg. Practicing a series of exercises for scoring high marks. Set, involves the readiness to act upon a series of challenges to overcome them. It includes the ability to imitate a displayed behavior or utilize a trial and error method to resolve a situation. Mechanism involves the ability to convert learned responses into habitual actions with proficiency and confidence. A student having an increased typing speed when using a computer is an example for complex overt response. Adaptability is an integral part of the domain which exhibits the ability to modify learned skills to meet special

events – making a working model. Origination also involves creating new movement patterns for a specific situation.

Cognitive is for mental skills, affective is for growth in feelings or emotional areas while psychomotor is manual or physical skills.

The learning domains can be incorporated, while designing the course outcomes of all the courses in a program. We can learn mental skills, develop our attitudes and acquire new physical skills as we perform the activities of our daily living.

Research in the field establishes the significant relation among three domain of learning.

A lesson developed by a teacher requires the inclusion of all three domains in constructing learning tasks for students. The diversity in such learning tasks helps creates a comparatively well-rounded learning experience that meets a number of learning styles and learning modalities. It is important for teachers to adopt a teaching strategy that combines three domains of learning to enable teaching and learning to be considered as effective. Apply the best and suitable delivery techniques that would impact positively on the Cognitive, Affective and Psychomotor domains of the students. The interaction of all three domains is needed for meaningful educational experiences.

For example while teaching practical geometry, New knowledge is acquired in Cognitive Domain. Using motor skills we teach how do the best in Psychomotor Domain. With the Affective Domain components we give motivation, praise to students to bring interest in the subject matter. In the same way we relate two or three Domains and perform teaching -learning activities and give meaningful learning experiences.

1.12 Correlation between Mathematics and other subjects

Mathematics and Science plays an important role in Correlation. It makes learning permanent and concrete and knowledge to the learner. It enhance the mental abilities like problem solving, logical reasoning, imagination and analytical power of student, because these can easily correlate acquired knowledge with the other subjects.

1. Mathematics and Physics

All laws and principles in Physics and expressed as equations using Mathematical language and symbols.

$$V = u + at$$

$$V^2 = u^2 + 2as$$

$$S = ut + \frac{1}{2}at^2$$

} laws of motion

$$T = 2\pi\sqrt{\frac{l}{g}}$$

oscillation time of the pendulum

$$F = ma$$

Newton's second law of motion

$$KE = \frac{1}{2}MV^2$$

Formula for kinetic energy

$$PV = RT$$

Bayles Law

$$\mu = \frac{\sin i}{\sin r}$$

light deflection number

Vermier Caliper and Screw Gauge which are used for accurate measurement need Mathematics knowledge. Modern war fare completely depend on the use of Mathematics knowledge. To lead the correct path of the missiles, Artificial Satellites and space crafts, we use the mathematics knowledge. We need the mathematics knowledge for higher education and to do research in physics topics. In the field of heath care all modern instruments are built with the knowledge of Mathematics.

2. Mathematics and Chemistry

All chemical combinations are governed by certain Mathematical law. In Chemical reactions equations are balance by balancing the number of atoms on either side of the equations. Balancing the chemical equations we need the knowledge of Mathematics. For example to prepare Chlorine gas in laboratory, heat with manganese die oxide with concentrate Hydrochloric acid. We can explain this with the help of equation $MnO_2 + 4 HCl \xrightarrow{\Delta} MnCl_2 + 2H_2O + Cl_2$. Again we know Hydrogen and Oxygen gives water. If we write $H_2 + O_2 \rightarrow H_2O$, then the equation is not balanced. We write the above equation as $2H_2 + O_2 \rightarrow 2H_2O$. From the above equation we realize 2 quote of Hydrogen and 1 quota of Oxygen gives two quota of water. Mixture of gases in air is expressed interms of percentage

Gases in air	Percentage
Oxygen	21%
Nitrogen	78%
Carbon dioxide, Argan, Neon etc.	1%

Instead of telling chemical reactions page by page, simple chemical equations with Mathematics knowledge gives in very short.

3. Mathematics and Biology

To study natural sciences it is practically impossible without the knowledge of higher Mathematical methods. The Indian Education Commission has also emphasized the importance of Mathematics in the development of Biological science in the following lines. "Apart from the Mathematics role in the growth of physical sciences it is now playing an increasingly important part in the development of biological sciences". Thus Biology and Medicine which seemed to have practically no use for Mathematics at the beginning of the last century, are benefiting immensely from the sophisticated mathematical tools. Statistical methods and

techniques are widely used for analysing complex problems relating to Physiology, genetics, heredity, metabolism and soon. Bacterial growth is expressed as an exponential function. Rate of Respiration, Normal weight with reference to height (BMI), caloric value of food is calculated using Maths.

4. Mathematics and social science

History, Civics, Psychology and many other subjects are mathematically oriented. Statistical analysis is the only scientific method to deal with social and psychological phenomena. The history of mathematics and their contributions provide useful background for the teaching of mathematics and history. Mathematics helps in calculating the dates and days of various historical events and in predicting the future based on the past events. Mathematics helps in drawing the maps to the scale and locating the distance between places. Mathematics provides the basis for computing longitude and latitude and planning the transportation system and so on.

5. Mathematics and Geography

Geographical figures are explained in the terms of numbers like seasonal conditions, temperature, rainfall, humidity etc. Mathematics is used to draw maps, formation of Days and Nights, solar and lunar eclipse, Barometric pressure, height above sea level. Calculation of international, local and standard time etc. Geography has gained a great deal in quantitative value and precision in adopting mathematical techniques. Mathematics provides the basis for computing longitude and latitude.

6. Mathematics and Economics

Statistical methods are used to calculate the volume of Trade, Trend of import and exports, Economic forecasts, Trade cycles, sensex index, crop production, population inflation etc. Diagrams like pictograph, Bar Diagram, Pie Diagram, Histogram Frequency curve, Ogive curve etc. are

frequently used in Business Statistics to represent data or to forecast. To draw all the above diagrams sound Mathematics knowledge is required. Mathematical concepts Average, Percentage, Variable, Constant etc. are used in Economics. Mathematical Topics such as Calculas, Matrices etc. are used in Economics. A great deal of mathematical thinking goes into the task of national economic planning.

7. Mathematics and language subjects

Mathematics help the students to learn the language with clarity and exactness. Basic arithmetic can enable students to better understand poetry. The length of poetry, the number of words to include in a line and the effect of certain rhythms are all product of mathematics. The linear, logical thinking used in mathematical problems can also help students write more clearly and logically. Abstract thinking and imagination used in Mathematics help students to write poems and short stories.

8. Mathematics and Fine Arts

A mathematical mind can appreciate art with some sense of confidence. Art, Sculpture thrives on geometry and students who understand congruent, identical, ratio, parallel etc. can bring lively beautiful art pieces. Musical rhythm often follows complex mathematical series and maths can help students to learn the basic rhythms of dances. Mathematics genius pythogorse applied mathematical theories for compose music.

Uses of correlation

1. Mind perceives knowledge as a whole. All the subjects aim to develop the intellectual power of the students as a whole. Therefore, it becomes necessary to correlate one subject with another. No subject can be taught in isolation.
2. Correlation of a subject with daily life helps to make the knowledge broad-based and makes knowledge useful.

The aim of education "to achieve the development of an all-round personality", cannot be done by teaching only a few subjects in isolation.

3. It strengthens skills that students encounter in one content area but also practice in another, leading to mastery of those skills.
4. It develops the mental abilities like imagination power, logical thinking and analytical power of students.
5. It makes the lesson easy and clear and develops various human and social qualities such as co-operation, citizenship etc.
6. Correlation of subject with daily life is of the utmost importance in order to create interest in the subject.
7. It makes learning concrete and permanent.
8. It helps to lighten the burden of curriculum.

Review Question

1. Explain the meaning, nature and scope of Mathematics.
2. What are the important Aims of teaching Mathematics in Schools?
3. Write the different values of Mathematics.
4. Explain the importance of Instructional objectives and Behavioural objectives in teaching.
5. Explain the Cognitive Domain and write its Instructional Objectives.
6. Explain the Affective Domain and write its Instructional Objectives.
7. Explain the Psychomotor Domain and write its Instructional Objectives.
8. Explain Revised Bloom's of Taxonomy.
9. How learning domains can be inter-related in teaching of Mathematics.
10. Explain how Mathematics is correlated with other subjects.